

Mobile Application Development

Practical No. 20

Develop a program to implement services like bluetooth/wifi.

Bluetooth:-

1. activity_main.xml :-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/b1"
        android:text="Bluetooth ON"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/b2"
        android:text="Bluetooth OFF"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

2. MainActivity.java

```
package com.example.myapplication;

import android.Manifest;
import android.bluetooth.BluetoothAdapter;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

public class MainActivity extends AppCompatActivity {

    BluetoothAdapter bt;
    Button on, off;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        on = findViewById(R.id.b1);
        off = findViewById(R.id.b2);

        bt = BluetoothAdapter.getDefaultAdapter();

        // Request permission for Android 12+
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
            if (ActivityCompat.checkSelfPermission(this,
                Manifest.permission.BLUETOOTH_CONNECT) !=
PackageManager.PERMISSION_GRANTED) {

                ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.BLUETOOTH_CONNECT},
1);
            }
        }

        // Bluetooth ON
        on.setOnClickListener(v -> {
            if (bt == null) {
                Toast.makeText(this, "Bluetooth not supported",
Toast.LENGTH_SHORT).show();
            } else if (!bt.isEnabled()) {
                Intent intent = new
Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
                startActivity(intent);
            } else {
                Toast.makeText(this, "Already ON", Toast.LENGTH_SHORT).show();
            }
        });

        // Bluetooth OFF
        off.setOnClickListener(v -> {
            if (bt != null && bt.isEnabled()) {
                bt.disable();
                Toast.makeText(this, "Bluetooth OFF",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}

```

3. AndroidManifest.xml

```

<uses-permission android:name="android.permission.BLUETOOTH"/>
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN"/>
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT"/>

```

- **Output:-**

